

VANESSA NOELYN KALAKI

Software Engineering Student | AUCA, Kigali, Rwanda

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PROFILE

Software Engineering student at AUCA with hands-on experience across the development lifecycle — from secure coding and penetration testing to data pipelines and applied machine learning. Combines a solid academic foundation in software engineering, big data, and cybersecurity with field experience in structured data collection and quality control. Comfortable working across Python, Java, and SQL, and quick to pick up new tools and languages.

EDUCATION

Bachelor's Degree in Software Engineering

2024 – 2028

Adventist University of Central Africa (AUCA), Kigali, Rwanda

A-Level Certificate of Education

2019 – 2022

Fawe Girls School, Kigali, Rwanda

O-Level Certificate

2017 – 2018

New Explorers Girls Academy, Kigali, Rwanda

TECHNICAL SKILLS

Programming: Python, Java, SQL

Data & Tools: PostgreSQL, Git, Tableau, Jupyter Notebook, Linux Administration

Practices: Software testing (ISTQB black-box techniques), SOLID principles, design patterns, penetration testing fundamentals

Productivity: Microsoft Word, PowerPoint, Excel, Access, internet-based research

CERTIFICATIONS

- NetOps Certification (x2)
- Machine Learning Certificate

ACADEMIC PROJECTS

- Big Data ETL Pipeline** — *AUCA Coursework* 2026
- Built a Python ETL pipeline against the WHO Global Health Observatory API, loading immunization-coverage data into PostgreSQL and validating it with pytest, then visualized results in a public Tableau dashboard.
- Security Assessment & Penetration Testing** — *AUCA Coursework* 2026
- Set up an isolated lab environment and carried out penetration testing engagements against vulnerable machines, documenting findings in Cyber Kill Chain-structured reports.
- Event-Driven Notification System** — *AUCA Coursework* 2026
- Designed and implemented a notification system in Java applying five behavioral design patterns (Chain of Responsibility, Command, Observer, State, Strategy).
- Hospital Appointment Management System** — *AUCA Coursework* 2026
- Developed a Java application demonstrating SOLID principles for managing hospital appointments end to end.

WORK & TEACHING EXPERIENCE

- Data Collector** — *International Centre for Tropical Agriculture (CIAT)* May 2023 – May 2024
- Conducted field surveys and accurately recorded respondent data, maintaining error-free, verified records.
 - Explained survey purposes to respondents and addressed their questions while following ethical data-collection guidelines and ensuring confidentiality.
 - Prepared and submitted summary reports, travelling to assigned field locations.
- Educational Content Creator** — *Rwanda Basic Education Board Curriculum* Ongoing
- Created interactive digital study guides and printable assessment materials for Senior 2 and Senior 3 mathematics, aligned to the national curriculum.

LANGUAGES

Spoken: English (excellent), Kinyarwanda (good), currently learning French and Italian

REFEREE

Mukagatera Alphonsia

Accountant, International Centre for Tropical Agriculture (CIAT)
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